CLASS 10 MATHEMATICS • CHAPTER 7

Coordinate Geometry

Exhaustive point-by-point presentation covering distance formula, section formula, mid-point applications, and solved examples directly from the textbook.

🎓 Akash Srivastva (www.akashmaths.online)

Introduction & Real-World Links

What is Coordinate Geometry?

- Algebraic tool for studying geometry of figures.
- Links algebra and geometry together: points are represented by coordinates (x, y).
- Abscissa (x-coordinate) and ordinate (y-coordinate).

Field Applications

- Widely applied in physics, engineering, navigation, seismology, and art.
- Enables us to compute precise distances between locations and analyze paths.

Section 7.1 Live Check

Q1. What do we call the x-coordinate and y-coordinate of a point respectively?

Q2. What is the general form of the coordinates of any point lying on the x-axis? On the y-axis?

Q3. How does coordinate geometry bridge algebra and geometry?

Deriving the Distance Formula

The Distance Formula

The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by applying Pythagoras theorem on coordinate differences:

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Special Case: Distance from Origin

The distance of any point $P(x, y)$ from the origin $O(0, 0)$ is simply $OP = \sqrt{x^2 + y^2}$.

Section 7.2 Live Check

Q1. State the distance formula between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$.

Q2. Which geometric theorem is fundamental in deriving the distance formula?

Q3. Why do we take only the positive square root value when computing coordinate distances?

The Internal Section Formula

Section Formula (Internal Division)

Coordinates of point P(x, y) dividing the line segment joining A(x_1, y_1) and B(x_2, y_2) internally in the ratio $m_1 : m_2$:

$$P(x, y) = \left(\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \frac{m_1y_2 + m_2y_1}{m_1 + m_2} \right)$$

Mid-Point Formula (Ratio 1:1)

Mid-point M of join of A(x_1, y_1) and B(x_2, y_2) is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

Section 7.3 Live Check

Q1. State the section formula for internal division of a line segment in ratio $m_1 : m_2$.

Q2. What is the simplified mid-point formula when the ratio is 1 : 1?

Q3. If ratio is expressed as k : 1, what do the section formula coordinates become?

Worked Exercise Highlights

Problem 1: Vertices of a Square (Ex 7.1/Ex 2)

- Check A(1, 7), B(4, 2), C(-1, -1), D(-4, 4).
- All four side lengths computed via distance formula are equal (= √34).
- Diagonals AC and BD are equal (= √68).
- **Conclusion: ABCD is a square.**

Problem 2: Division Ratio (Ex 7.2/Ex 7)

- Find ratio in which point (-4, 6) divides A(-6, 10) and B(3, -8).
- Let ratio be k : 1 or $m_1 : m_2$.
- Equating x-coordinate gives $7m_1 = 2m_2$.
- **Ratio = 2 : 7 internally.**

Exercise Highlights Live Check

Q1. What two geometric conditions must be satisfied to prove a quadrilateral is a square using the distance formula?

Q2. How do we find an unknown ratio when dividing a segment given the dividing point's coordinates?

Q3. What is trisection of a line segment, and what ratios are used to find the two points?

Chapter Recap & Checklist

Distance Foundations

- **Distance Formula:** $d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$.
- **From Origin:** $OP = \sqrt{(x^2 + y^2)}$.

Division Formulas

- **Section Formula:** $((m_1x_2 + m_2x_1) / (m_1 + m_2), (m_1y_2 + m_2y_1) / (m_1 + m_2))$.
- **Mid-Point Formula:** $((x_1 + x_2) / 2, (y_1 + y_2) / 2)$.

THANK YOU

"Geometry is the art of correct reasoning on incorrect figures." — George Pólya

🎓 Akash Srivastva (www.akashmaths.online)